

SCHEME OF TEACHING IV SEMESTER

B.E. in Computer Science and Engineering (AI & ML)
Scheme of Teaching Batch - 2024

IV Semester

Sl. No.	Subject Code	Subject	Teaching Department	Category	Credits				Total contact hours/week			
					L	T	P	Total	L	T	P	S
1	CI41	Statistics, Probability And Linear Programming	Maths	ASC	2	1	0	3	2	2	0	2
2	CI42	Operating Systems	CSE(AI ML)	IC	3	0	1	4	3	0	2	3
3	CI43	Design and Analysis of Algorithms	CSE(AI ML)	PCC	3	0	0	3	3	0	0	3
4	CI44	Automata Theory and Compiler Design	CSE(AI ML)	PCC	3	0	0	3	3	0	0	3
5	CI45	Database Management Systems	CSE(AI ML)	PCC	3	0	0	3	3	0	0	3
6	CIL46	Database Management Systems Laboratory	CSE(AI ML)	PCC	0	0	1	1	0	0	2	0
7	CIL47	Algorithms Laboratory	CSE(AI ML)	PCC	0	0	1	1	0	0	2	0
8	CIL48	Data Visualization Laboratory	CSE(AI ML)	PCC	0	0	1	1	0	0	2	0
9	CIAEC49	Ability Enhancement Course – IV NoSQL Databases	CSE(AI ML)	AEC	1	0	0	1	1	0	0	1
Total					15	1	4	20	15	2	8	15
10	INT410	Inter/ Intra Institutional Internship		NCMC	0	0	0	0	-			
11	AM41	Additional Mathematics II*		NCMC	0	0	0	0	-			

NOTE: Minimum of 1 subject should have a Tutorial component of 1 Credit.

Nomenclature: **ASC:** Applied Science Course, **IPCC:** Integrated Professional Core Course, **PCC:** Professional Core Course, **INT** –Internship, **HSMC:** Humanity and Social Science & Management Courses, **AEC**–Ability Enhancement Courses, **NMC:** Non-credit Mandatory Course

L –Lecture, T – Tutorial, P- Practical/ Drawing, S-Self Learning

Integrated Professional Core Course (IPCC): Refers to Professional Theory Core Course Integrated with practical of the same course. Credit for IPCC is 03 and its Teaching–Learning hours (L : T : P: S) can be considered as (2 : 0 : 1). The theory part of the IPCC shall be evaluated both by CIE and SEE. The practical part shall be evaluated only by CIE (no SEE). However, questions from the practical part of IPCC can be included in the SEE question paper.

Innovation/ Societal/ Entrepreneurship based Internship: At the End of fourth Semester four - weeks summer internship shall be carried out at industry, State and Central Govt./NGO/MSME, Innovation centres or incubation centres. The internship can be Rural Internship. All the students shall have to undergo mandatory internship of 04 weeks during the intervening period of IV & V semesters. A Viva-Voce examination (CIE) shall be conducted during V semester and the prescribed credit shall be included in VI semester. Internship shall be considered as a head of passing and shall be considered for the award of degree. Those, who do not take-up/complete the internship shall be declared fail and shall have to complete during subsequent examination after satisfying the internship requirements. The in-charge faculty has to monitor the student’s internship progress and interact to guide them for the successful completion of the internship.

Innovation/ Societal/ Entrepreneurship based Internship shall have only CIE, no SEE component.

*** Lateral Entry Students:**

The Non-Credit Mandatory Course, Additional Mathematics II is prescribed for IV Semester Lateral Entry Diploma students admitted to III Semester of BE Program. The student shall register for this course along with other IV semester courses. The students shall attend classes for the course during the semester and complete all formalities of attendance and CIE. In case, any student fails to secure the minimum 40% of the prescribed CIE marks, he/she shall be deemed to have secured an F grade. In such a case, the student has to fulfill the requirements during subsequent semester/s to appear for CIE. In case student fails to register for the said course/ falls short of attendance, he/she will repeat the course whenever it is offered next. Additional Mathematics II shall have CIE component only and no SEE component. This Course shall not be considered for vertical progression, but completion of the course shall be mandatory for the award of the degree.

AICTE Activity Points to be earned by students admitted to BE program (For more details refer to Chapter 6, AICTE, Activity Point Program, Model Internship Guidelines):

Every regular student, who is admitted to the 4-year degree program, is required to earn 100 activity points in addition to the total credits earned for the program. Students entering 4 years' degree program through lateral entry are required to earn 75 activity points in addition to the total credits earned for the program. The activity points earned by the student shall be reflected on the students' VIII semester grade card. The activities to earn the points can be spread over the duration of the course. However, minimum prescribed duration should be fulfilled. Activity points (non-credit) have no effect on SGPA/CGPA and shall not be considered for vertical progression. In case student fail to earn the prescribed activity points; VIII semester grade card shall be issued only after earning the required activity Points. Students shall be eligible for the award of degree only after the release of the VIII semester grade card.

Semester IV

STATISTICS, PROBABILITY AND LINEAR PROGRAMMING	
Course Code: CI41	Credits: 2:1:0:2
Pre – requisites: Basic Probability	Contact Hours: 30L+15T+30L
Course Coordinator: Dr. Govindaraju M V	

Unit I

Statistics: Curve fitting by the method of least squares, fitting linear, quadratic and geometric curves, correlation, regression and multiple regression.

Probability Distributions: Random variables, Binomial distribution, Poisson distribution.

- Pedagogy/Course delivery tools: Chalk and talk, PowerPoint Presentation, Videos
- Links: <https://nptel.ac.in/courses/111/105/111105035/>
- Impartus recording: <https://a.impartus.com/ilc/#/course/96127/452>
<https://a.impartus.com/ilc/#/course/132243/636>
<https://a.impartus.com/ilc/#/course/119635/593>

Unit II

Probability Distributions: Uniform distribution, Exponential distribution, Gamma distribution and Normal distribution.

Joint probability distribution: Joint probability distribution (both discrete and continuous), conditional probability, conditional expectation.

- Pedagogy/Course delivery tools: Chalk and talk, PowerPoint Presentation, Videos
- Links: <https://nptel.ac.in/courses/111/105/111105035/>
<https://nptel.ac.in/courses/111/107/111107119/>
- Impartus recording: <https://a.impartus.com/ilc/#/course/96127/452>
<https://a.impartus.com/ilc/#/course/119635/593>

Unit III

Markov Chain: Introduction to stochastic process, probability vectors, stochastic matrices, regular stochastic matrices, Markov chains, higher transition probabilities, stationary distribution of regular Markov chains and absorbing states, Markov and Poisson processes.

Queuing theory: Introduction, symbolic representation of a queuing model, single server Poisson queuing model with infinite capacity (M/M/1 :FIFO), single server Poisson queuing model with finite capacity (M/M/S : N/FIFO), multiple server Poisson queuing model with infinite capacity (M/M/S :FIFO), Multiple server

Poisson queuing model with finite capacity (M/M/S : N/FIFO), introduction to M/G/1 queuing model.

- Pedagogy/Course delivery tools: Chalk and talk, PowerPoint Presentation, Videos
- Links: <https://nptel.ac.in/courses/111103022>
- Impartus recording: <https://a.impartus.com/ilc/#/course/96127/452>
<https://a.impartus.com/ilc/#/course/119635/593>

Unit IV

Sampling and Statistical Inference: Sampling distributions, central limit theorem, concepts of standard error and confidence interval, level of significance, type I and type II errors, one tailed and two tailed tests, Z-test: for single mean, for single proportion and for difference between means, Student's t –test: for single mean and for difference between two means, F – test for equality of two variances, Chi-square test: for goodness of fit and for independence of attributes.

- Pedagogy/Course delivery tools: Chalk and talk, PowerPoint Presentation, Videos
- Links: <https://nptel.ac.in/courses/111/105/111105035/>
<https://nptel.ac.in/courses/111/107/111107119/>
- Impartus recording: <https://a.impartus.com/ilc/#/course/96127/452>
<https://a.impartus.com/ilc/#/course/119635/593>

Unit V

Linear Programming: Introduction to linear programming problem (LPP), formulation of the problem, graphical method, general, canonical and standard forms of LPP, simplex method, big-M method, two-phase simplex method and duality in linear programming.

- Pedagogy/Course delivery tools: Chalk and talk, PowerPoint Presentation, Videos
- Links: <https://nptel.ac.in/courses/111104027>
- Impartus recording: <https://a.impartus.com/ilc/#/course/96127/452>
<https://a.impartus.com/ilc/#/course/119635/593>

Text Books:

1. R.E. Walpole, R. H. Myers, R. S. L. Myers and K. Ye – Probability and Statistics for Engineers and Scientists – Pearson Education – Delhi – 9th edition – 2012.
2. B.S.Grewal - Higher Engineering Mathematics - Khanna Publishers – 44th edition-2017.

3. T. Veerarajan- Probability, Statistics and Random processes – Tata McGraw-Hill Education – 3rd edition -2017.
4. Kanti Swarup, P.K. Gupta and Man Mohan -Operations Research-Sultan Chand & Sons Publishers–2014.

Reference Books:

1. Erwin Kreyszig - Advanced Engineering Mathematics-Wiley-India publishers-10th edition-2015.
2. Sheldon M. Ross – Probability models for Computer Science – Academic Press, Elsevier– 2009.
3. Murray R Spiegel, John Schiller & R. Alu Srinivasan – Probability and Statistics – Schaum’s outlines -4th edition-2012.
4. Kishore S. Trivedi – Probability & Statistics with Reliability, Queuing and Computer Science Applications – John Wiley & Sons – 2nd edition – 2008.
5. Johnson/Miller: Miller & Freund's Probability and Statistics for Engineers , Eighth Edition, Pearson Education India -2015

Course Outcomes

At the end of the course, students will be able to

1. Fit a least squares curve to the given data and analyze the given random data and its probability distributions. (PO-1, 2, PSO-2, 3)
2. Find parameters of Continuous Probability distributions and calculate the marginal and conditional distributions of bivariate random variables. (PO-1, 2, PSO-2, 3)
3. Predict future events using Markov chain and in queuing models. (PO-1, 2, PSO-2, 3)
4. Choose an appropriate test of significance and make inference about the population from a sample. (PO-1, 2, PSO-2, 3)
5. Formulate and solve a simple linear programming problem. (PO-1, 2, PSO-1, 3)

Course Assessment and Evaluation

Continuous Internal Evaluation: 50 Marks		
Assessment Tool	Marks	Course outcomes addressed
Internal test-I	30	CO1, CO2, CO3
Internal test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other components	Marks	Course outcomes addressed
Quiz	10	CO1, CO2, CO3
Assignment	10	CO3, CO4, CO5
Semester End Examination:	100	CO1, CO2, CO3, CO4, CO5

OPERATING SYSTEMS	
Course Code: CI42	Credits: 3:0:1:3
Pre – requisites: nil	Contact Hours: 45L+15P+45S
Course Coordinator: Mrs. Akshatha G C	

Course Contents

Unit I

Introduction: What operating systems do; Computer System organization; Computer System architecture; Operating System structure; Operating System operations;

Operating system structures: operating system services, user operating system Interface, System calls, Types of system calls, Operating system structure, System boot

- Pedagogy/Course delivery tools: Chalk and talk, PowerPoint Presentation, Videos
- Links: <https://nptel.ac.in/courses/106105214>
- Impartus recording: <https://a.impartus.com/ilc/#/course/148805/703>

Unit II

Process Management: Basic concept; Process scheduling; Operations on processes; Inter process Communication.

Threads: Overview; Multithreading models;

Process scheduling: Basic concepts, Scheduling criteria, scheduling algorithms, multiple processor scheduling, Algorithm evaluation.

- Pedagogy/Course delivery tools: Chalk and talk, PowerPoint Presentation, Videos
- Links: <https://nptel.ac.in/courses/106105214>
- Impartus recording: <https://a.impartus.com/ilc/#/course/148805/703>

Unit III

Process Synchronization: Synchronization, The Critical section problem; Peterson's solution; Synchronization hardware; Semaphores; Classical problems of synchronization; Monitors. **Deadlocks:** System model; Deadlock characterization; Methods for handling deadlocks; Deadlock prevention; Deadlock avoidance; Deadlock detection and recovery from deadlock.

- Pedagogy/Course delivery tools: Chalk and talk, PowerPoint Presentation, Videos
- Links: <https://nptel.ac.in/courses/106105214>
- Impartus recording: <https://a.impartus.com/ilc/#/course/148805/703>

Unit IV

Memory Management Strategies: Background; Swapping; Contiguous memory allocation; Paging; Structure of page table; Segmentation.

Virtual Memory Management: Background; Demand paging; Copy-on write; Page replacement; Allocation of frames; Thrashing.

- Pedagogy/Course delivery tools: Chalk and talk, PowerPoint Presentation, Videos
- Links: <https://nptel.ac.in/courses/106105214>
- Impartus recording: <https://a.impartus.com/ilc/#/course/148805/703>

Unit V

File System: File concept; Access methods; Directory structure; File system mounting; file sharing; protection.

Secondary Storage Structures: Disk scheduling; FCFS Scheduling, SSTF scheduling, SCAN, C-SCAN scheduling, Look Scheduling, CLOOK scheduling.

System Protection: Goals of protection, Principles of protection, Domain of protection, Access matrix.

- Pedagogy/Course delivery tools: Chalk and talk, PowerPoint Presentation, Videos
- Links: <https://nptel.ac.in/courses/106105214>
- Impartus recording: <https://a.impartus.com/ilc/#/course/148805/703>

Lab Experiments:

Students will execute the following contents as part of practical component:

1. Study of Unix/Linux general purpose utility command list: man, who, cat, cd, cp, ps, ls, mv, rm, mkdir, rmdir, echo, more, date, time, kill, history, chmod, chown, finger, pwd, cal, logout, shutdown.
2. Write a c/c++ program that outputs the contents of its environment list.
3. Write a c/c++ program to check the following limit:
 - a. No of clock ticks
 - b. Maximum number of child process
 - c. Maximum path length

- d. Maximum no of characters in a filename
 - e. Maximum no of open files/processes
4. Write a C program to create a child process and allow the parent to display “parent” and the child to execute copy command to copy the content of the file specified as command line argument while executing the program.
 5. Write a C/C++ program to fetch the termination status from the child and display whether it terminated normally or abnormally.
 6. Write C/C++ program that illustrate communication between two unrelated processes using named pipe (FIFO file).
 7. Write a c program to demonstrate and avoid race condition.
 8. Write a C/C++ program to demonstrate FCFS/SJF/SRTF and priority CPU scheduling policies.
 9. Write a c program to demonstrate synchronization between producer consumer processes that share a common buffer and performs a read and write operation to and from the buffer. Propose a solution that satisfies all the three requirements of the solution to critical section problem.
 10. Write a C/C++ program to query and display the different attributes associated with a file.
 11. Write a C/C++ program to implement banker’s algorithm.
 12. Consider the last 100 bytes as a region. Write a C/C++ program to check whether the region is locked or not. If the region is locked, print pid of the process which has locked. If the region is not locked, lock the region with an exclusive lock, read the last 50 bytes and unlock the region.
 13. Write a program to implement FIFO, LRU and optimal page replacement algorithms.
 14. Write a C/C++ program to implement user defined cat, ls commands which works similar to built-in cat and ls commands.

Suggested Learning Resources

Text Books:

1. Abraham Silberschatz, Peter Baer Galvin, Greg Gagne Operating System Principles, 8th edition Wiley- India,2011.

Reference Books:

1. D.M Dhamdhare Operating systems - A concept-based Approach, 2nd Edition, Tata McGraw-Hill, 2002
2. Harvey M Deital Operating systems, 3rd Edition, Addison Wesley, 1990.
3. Operating Systems: Principles and Practice (2nd Edition), by Thomas Anderson and Michael Dahlin.

Course Outcomes (COs):

At the end of the course students will be able to:

1. Describe the structure of the computer system and services provided by the Operating system. (PO-1,2, PSO-1,2)
2. Apply different scheduling algorithms for Process/Memory/Disk Management. (PO-1,2,3,4, PSO-1,2)
3. Describe Process management and need for controlled access to computing resources by co-operative processes. (PO-1,2,3,4, PSO-1,2)
4. Apply deadlock detection and prevention algorithms to solve the given problem. (PO-1,2,3,4, PSO-1,2,3)
5. Illustrate memory management strategies and operating system principles for achieving protection and security. (PO-1,2,3,4, PSO-1,2)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment Tools	Marks	Course Outcomes (COs) addressed
Internal Test-I (CIE-I)	30	CO1, CO2
Internal Test-II (CIE-II)	30	CO3, CO4, CO5
Average of the two CIE shall be taken for 30 marks		
Other Components		
Lab Test	10	CO1, CO2, CO3, CO4, CO5
Lab Record	10	CO1, CO2, CO3, CO4, CO5
The Final CIE out of 50 Marks = Average of two CIE tests for 30 Marks+ Marks scored in Quiz + Marks scored in Assignment		
Semester End Examination (SEE)	100	CO1, CO2, CO3 CO4, CO5

DESIGN AND ANALYSIS OF ALGORITHMS	
Course Code: CI43	Credits: 3:0:0:3
Pre – requisites: Nil	Contact Hours: 45L+45S
Course Coordinator: Dr. Sini Anna Alex	

Course Contents

Unit I

Asymptotic Bounds and Representation problems of Algorithms: Computational Tractability: Some Initial Attempts at Defining Efficiency, Worst-Case Running Times and Brute-Force Search, Polynomial Time as a Definition of Efficiency, Asymptotic Order of Growth: Properties of Asymptotic Growth Rates, Asymptotic Bounds for Some Common Functions, A Survey of Common Running Times: Linear Time, $O(n \log n)$ Time, $O(n^k)$ Time, Beyond Polynomial Time. Substitution Method, Some Representative Problems, A First Problem: Stable Matching. (Textbook 1)

- Pedagogy/Course delivery tools: Chalk and talk, PowerPoint Presentation, Videos
- Links: <https://nptel.ac.in/courses/106106131>
<https://nptel.ac.in/courses/106102064>

Unit II

Graphs & Divide and Conquer: Graph Connectivity and Graph Traversal, Breadth-First Search: Exploring a Connected Component, Depth-First Search, Implementing Graph Traversal Using Queues and Stacks: Implementing Breadth-First Search, Implementing Depth-First Search, An Application of Breadth-First Search and Depth-First Search, Directed Acyclic Graphs and Topological Ordering. Divide and Conquer Technique: Masters Theorem for recurrence relations, The Merge sort Algorithm, Quick Sort Algorithm. (Textbook 2)

- Pedagogy/Course delivery tools: Chalk and talk, PowerPoint Presentation, Videos
- Links: <https://nptel.ac.in/courses/106106131>
<https://nptel.ac.in/courses/106102064>
<https://www.geeksforgeeks.org/fundamentals-ofalgorithms/?ref=shm>

Unit III

Transform and Conquer: Heaps and Heapsort. (Textbook 2) **Greedy Algorithms:** Interval Scheduling: The Greedy Algorithm Stays Ahead: Designing a Greedy

Algorithm, Analyzing the Algorithm, Scheduling to Minimize Lateness: An Exchange Argument: The Problem, Designing the Algorithm. (Textbook 1)
Prim's Algorithm, Kruskal's Algorithm, Dijkstra's Algorithm, Huffman Trees and Codes. (Textbook 2)

- Pedagogy/Course delivery tools: Chalk and talk, PowerPoint Presentation, Videos
- Links: <https://nptel.ac.in/courses/106106131>
<https://nptel.ac.in/courses/106102064>
<https://www.geeksforgeeks.org/fundamentals-of-algorithms/?ref=shm>

Unit IV

Dynamic Programming: Weighted Interval Scheduling: A Recursive Procedure, Subset Sums and Knapsacks: Adding a Variable: The Problem, Designing the Algorithm. (Textbook 1)

Dynamic Programming: Warshall's and Floyd's Algorithm. (Textbook 2)

Iterative Improvement: The Simplex Method, The Maximum-Flow Problem. (Textbook 2)

- Pedagogy/Course delivery tools: Chalk and talk, PowerPoint Presentation, Videos
- Links: <https://nptel.ac.in/courses/106106131>
<https://nptel.ac.in/courses/106102064>
<https://www.geeksforgeeks.org/fundamentals-of-algorithms/?ref=shm>

Unit V

NP and Computational Intractability: Polynomial-Time Reductions NP-Complete Problems: Circuit Satisfiability: A First NP-Complete Problem, General Strategy for Proving New Problems NP Complete, Sequencing Problems: The Traveling Salesman Problem, The Hamiltonian Cycle Problem.

- Pedagogy/Course delivery tools: Chalk and talk, PowerPoint Presentation, Videos
- Links: <https://www.geeksforgeeks.org/fundamentals-of-algorithms/?ref=shm>

Suggested Learning Resources

Text Books:

1. Algorithm Design, Jon Kleinberg and Eva Tardos, Pearson, 1st Edition 2013.
2. Introduction to the Design & Analysis of Algorithms, Anany Levitin, 3rd Edition, Paperback Publication -2017, Pearson education

Reference Books:

1. Design and Analysis of Algorithms, Michael. T. Goodrich, Roberto Tamassia, An Indian Adaptation, Wiley, 2023
2. Introduction to Algorithms, H., Cormen, Charles E. Leiserson, Ronal L. Rivest, Clifford Stein Thomas, 3rd Edition, 2009, MIT press.
3. Fundamentals of Computer Algorithms, Horowitz E., Sartaj Sahni S., Rajasekaran S, 2008, Galgotia Publications.

Course Outcomes (COs):

At the end of the course student will be able to:

1. Understand the basic concepts and analyse worst-case running times of algorithms using asymptotic analysis. (PO-1,2,3 PSO-1,2)
2. Analyze the design techniques for graph traversal using representative algorithms. (PO-1,2,3 PSO-1,2)
3. Demonstrate divide and conquer, transform and conquer techniques and analyse the complexity of the methods by solving recurrence. (PO-1,2,3 PSO-1,2)
4. Illustrate Greedy paradigm and Dynamic programming paradigm using representative algorithms. (PO-1,2,3,4 PSO-1,2,3)
5. Describe the classes P, NP, and NP-Complete and be able to prove that a certain problem is NP-Complete. (PO-1,2,3,4 PSO-1,2,3)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment Tools	Marks	Course Outcomes (COs) addressed
Internal Test-I (CIE-I)	30	CO1, CO2
Internal Test-II CIE-II)	30	CO3, CO4, CO5
Average of the two CIE shall be taken for 30 marks		
Other Components		
Case study	10	CO1, CO2, CO3
Assignment	10	CO4, CO5
The Final CIE out of 50 Marks = Average of two CIE tests for 30 Marks+ Marks scored in Assessment Test +Marks scored in Assignment		
Semester End Examination (SEE)	100	CO1, CO2, CO3, CO4, CO5

AUTOMATA THEORY AND COMPILER DESIGN	
Course Code: CI44	Credits: 3:0:0:3
Pre – requisites: Nil	Contact Hours: 45L+45S
Course Coordinator: Dr. Sini Anna Alex	

Course Contents

Unit I

Introduction to Automata Theory: Central Concepts of Automata theory, Deterministic Finite Automata (DFA), Non- Deterministic Finite Automata (NFA), NFA to DFA Conversion, Minimization of DFA. **Introduction to Compiler Design:** Language Processors, Phases of Compilers, Compiler Construction Tools.

- Pedagogy/Course delivery tools: Chalk and talk, PowerPoint Presentation, Videos
- Links: <https://nptel.ac.in/courses/106104148>
<https://nptel.ac.in/courses/106108113>

Unit II

Regular Expressions and Languages: Regular Expressions, Finite Automata and Regular Expressions. **Push Down Automata:** Definition of the Pushdown Automata, The Languages of a PDA. **Lexical analysis Phase of Compiler Design:** The Role of Lexical Analyzer, Input Buffering, Specifications of Tokens, Recognition of Tokens.

- Pedagogy/Course delivery tools: Chalk and talk, PowerPoint Presentation, Videos
- Links: <https://nptel.ac.in/courses/106104148>
<https://nptel.ac.in/courses/106108113>

Unit III

Context Free Grammars: Formal Definition of a Context Free Grammar, Derivations, Parse Trees, Ambiguity, writing a Grammar: Elimination of Ambiguity, Elimination of Left Recursion, Left Factoring.

Syntax Analysis: The role of the Parser, Top-down Parsing, First and Follow, LL (1) Grammars, Non-Recursive Predictive Parsing, Error Recovery in Predictive Parsing. Bottom-up Parsing, Introduction to LR Parsing: Simple LR parser. More Powerful LR Parsers: Canonical LR (1) items, Constructing LR (1) set of items, Canonical LR (1) parse tables, Constructing LALR parsing tables

- Pedagogy/Course delivery tools: Chalk and talk, PowerPoint Presentation, Videos

- Links: <https://nptel.ac.in/courses/106108113>

Unit IV

Introduction to Turing Machine: Problems that Computers Cannot Solve, The Turing machine.

Syntax-Directed Translation: Syntax-Directed Definitions, Evaluation order for SDDs, Applications of Syntax-Directed translation, Syntax-Directed Translation Schemes

- Pedagogy/Course delivery tools: Chalk and talk, PowerPoint Presentation, Videos
- Links: <https://nptel.ac.in/courses/106104148>
<https://nptel.ac.in/courses/106108113>

Unit V

Intermediate Code Generation: Variants of syntax trees, Three-address code, Types and Declarations, Translation of expressions, Type Checking: Rules of Type Checking, Type Conversions, Control flow: Boolean Expressions, Short-Circuit Code, Flow of Control statements, Control Flow Translation of Boolean Expressions, Avoiding Redundant Gotos, Back patching: One pass Code generation Using backpatching, Backpatching for Boolean Expressions.

Code Generation: Issues in the Design of a Code Generator, Target Language, A Simple Code Generator.

- Pedagogy/Course delivery tools: Chalk and talk, PowerPoint Presentation, Videos
- Links: <https://nptel.ac.in/courses/106108113>

Suggested Learning Resources

Text Books:

1. John E Hopcroft, Rajeev Motwani, Jeffrey D. Ullman, “Introduction to Automata Theory, Languages and Computation”, Third Edition, Pearson Education India, 2011.
2. Alfred V.Aho, Monica S.Lam, Ravi Sethi, Jeffrey D. Ullman, “ Compilers Principles, Techniques and Tools”, Second Edition, Pearson Education India, 2013

Reference Books:

1. Elain Rich, “Automata, Computability and complexity”, 1st Edition, Pearson Education, 2018.
2. K.L.P Mishra, N Chandrashekar , 3rd Edition , ‘Theory of Computer Science’, PHI, 2012.
3. Peter Linz, “An introduction to Formal Languages and Automata“, 3rd Edition, Narosa Publishers, 1998

4. K Muneeswaran,” Compiler Design”, Oxford University Press 2013

Course Outcomes (COs):

At the end of the course student will be able to:

1. Understand the core concepts in automata theory and compiler design. (PO-1,2, PSO-1,2)
2. Design and develop lexical analysers, parsers and code generators. (PO-1,2,PSO-1,2)
3. Design Grammars and Automata (recognizers) for different language classes and become knowledgeable about restricted models of Computation (Regular, Context Free) and their relative powers. (PO-1,2,3,4, PSO-1,2)
4. Demonstrate the structure of a Compiler and Apply the concepts of automata theory and Theory of Computation to design Compilers.(PO-1,2,3,4, PSO-1,2)
5. Design syntax directed computation models for problems in Automata theory and adaptation of such models in the field of compilers. (PO-1,2,3,4, PSO-1,2)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment Tool	Marks	Course outcomes addressed
Internal test-I	30	CO1, CO2, CO3
Internal test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other components		
Active Learning: Technical Paper Writing, Game Based Learning	10	CO1, CO2, CO3
Tutorial Assignment	10	CO1,CO2, CO3, CO4, CO5
Semester End Examination (SEE)	100	CO1, CO2, CO3, CO4, CO5

DATABASE MANAGEMENT SYSTEMS	
Course Code: CI45	Credits: 3:0:0:3
Pre – requisites: Nil	Contact Hours: 45L+45S
Course Coordinator: Dr. Sini Anna Alex	

Course Contents

Unit I

Introduction: Characteristics of Database approach, Actors on the Scene, Workers behind the scene, Advantages of using DBMS approach, Data models, schemas and instances, Three-schema architecture and data independence, Database languages and interfaces, the database system environment, Entity-Relationship Model: Conceptual Database using high-level Conceptual data models for Database Design, A Sample Database Application, Entity types, Entity sets Attributes and Keys Relationship types, Relationship Sets, Roles and Structural Constraints Weak Entity Types.

- Pedagogy/Course delivery tools: Chalk and talk, PowerPoint Presentation, Videos
- Links: <https://nptel.ac.in/courses/106105175>
<https://nptel.ac.in/courses/106106220>

Unit II

Relational Database: Design Using ER- to-Relational Mapping, Relational Model and Relational Algebra: Relational Model Concepts, Relational Model Concepts, Relational Model Constraints and Relational Database Schema Update Operations, Transactions and Dealing with Constraint violations, Unary Relational operations, Relational Algebra Operations from Set Theory, Binary Relational Operations, JOIN and DIVISION, Additional Relational Operations, Examples of Queries in Relational Algebra.

- Pedagogy/Course delivery tools: Chalk and talk, PowerPoint Presentation, Videos
- Links: <https://nptel.ac.in/courses/106105175>
<https://nptel.ac.in/courses/106106220>

Unit III

Introduction to SQL: Overview of the SQL Query Language, SQL Data Definition, Basic structure of SQL Queries, Additional Basic Operations, Null values, Aggregate Functions, nested Sub queries, Modification of the Database, Join Expressions, Views, Integrity Constraints, SQL Data Types and Schemas.

- Pedagogy/Course delivery tools: Chalk and talk, PowerPoint Presentation,

Videos

- Links: <https://nptel.ac.in/courses/106105175>
<https://nptel.ac.in/courses/106106220>

Unit IV

Database Design: Informal Design Guidelines for Relation Schemas, Functional Dependencies, Inference Rules, Equivalence and Minimal Cover, Normal Forms based on Primary Keys, First Normal Form, General Definitions of 2nd and 3rd Normal Forms, Boyce Codd Normal Form, Properties of Relational Decomposition, Algorithms for relational database schema design.

- Pedagogy/Course delivery tools: Chalk and talk, PowerPoint Presentation, Videos
- Links: <https://nptel.ac.in/courses/106105175>
<https://nptel.ac.in/courses/106106220>

Unit V

Transaction Management: Transaction Concept, a Simple Transaction Model, Transaction Atomicity and Durability, Serializability, Transaction Isolation and Atomicity, Transaction Isolation Levels. Concurrency Control: Lock-Based Protocols, Deadlock Handling. Recovery System: Failure Classification, Recovery and Atomicity, Recovery Algorithm.

- Pedagogy/Course delivery tools: Chalk and talk, PowerPoint Presentation, Videos
- Links: <https://nptel.ac.in/courses/106105175>
<https://nptel.ac.in/courses/106106220>

Suggested Learning Resources

Text Books:

1. Ramez Elmasri and Shamkant B. Navathe: Fundamentals of Database Systems, 7th Edition, Pearson, 2016.
2. Silberschatz, Korth and Sudharshan: Data base System Concepts, 7th Edition, Mc-GrawHill, 2021.

Reference Books:

1. C.J. Date, A. Kannan, S. Swamynatham, An Introduction to Database Systems, 8th Edition, Pearson Education, 2009.
2. Raghu Ramakrishnan and Johannes Gehrke: Database Management Systems, 3rd Edition, McGraw-Hill, 2003.

Course Outcomes (COs):

At the end of the course student will be able to:

1. Design entity-relationship diagrams to represent simple database applications and convert them to a Relational model. (PO-2, 3, 4, 5, PSO-2)
2. Construct relational algebraic expressions for queries using the concepts of relational database theory. (PO-1, 2, 4, PSO-2)
3. Formulate using SQL, solutions to a broad range of query and data update problems. (PO-1,2,3,4,5,7,8,9 ,PSO-1,2,3)
4. Apply Normalization to improve database design strategies. (PO-1, 2,3 PSO-2)
5. Interpret the basic issues of transaction processing, concurrency control, and recovery techniques. (PO-1,2,3, PSO-1,2)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment Tool	Marks	Course outcomes addressed
Internal test-I	30	CO1, CO2, CO3
Internal test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other components		
Case study	10	CO1, CO2, CO3
project	10	CO3, CO4, CO5
Semester End Examination (SEE)	100	CO1, CO2, CO3, CO4, CO5

DATABASE MANAGEMENT LABORATORY	
Course Code: CIL46	Credits: 0:0:1
Pre – requisites: Nil	Contact Hours: 15P
Course Coordinator: Dr. Sini Anna Alex	

Course Contents

Sl. No.	Lab Course Contents
1.	Construct Entity Relationship Diagram for the scenario given below Consider a movie database in which data is recorded about the movie industry. The data requirements are summarized as follows: - Each movie has a title, year of release, group of actors, one director and production company. Each movie has different genre. - It is required to keep track of the name, the address and the owner of the production company. - Data to be kept about directors and actors are name, SSN, date of birth, address, mobile phone number. An actor may also be a director.
2.	Construct Entity Relationship Diagram for the scenario given below. The data requirements are summarized as follows: Employees work in departments; each department is managed by an employee; a child must be identified uniquely by name when the parent (who is an employee; assume that only one parent works for the company) is known. We are not interested in information about a child once the parent leaves the company.
3.	Construct Entity Relationship Diagram for the scenario given below The data requirements are summarized as follows: - Galleries keep information about artists, their names (which are unique), birthplaces, age, and style of art. For each piece of artwork, the artist, the year it was made, its unique title, its type of art (e.g., painting, lithograph, sculpture, photograph), and its price must be stored. Pieces of artwork are also classified into groups of various kinds, for example, portraits, still lifes, works by Picasso, or works of the 19th century; a given piece may belong to more than one group. - Each group is identified by a name (like those just given) that

	describes the group. Finally, galleries keep information about customers. For each customer, galleries keep that person's unique name, address, total amount of dollars spent in the gallery (very important!), and the artists and groups of art that the customer tends to like.
4.	Convert ER to Relational Schema for the following scenarios - Consider a movie database in which data is recorded about the movie industry. The data requirements are summarized as follows: - Each movie has a title, year of release, group of actors, one director and production company. Each movie has different genre. - It is required to keep track of the name, the address and the owner of the production company. - Data to be kept about directors and actors are name, SSN, date of birth, address, mobile phone number. An actor may also be a director. - Construct the employee database given. The data requirements are summarized as follows: - Employees work in departments; each department is managed by an employee; a child must be identified uniquely by name when the parent (who is an employee; assume that only one parent works for the company) is known. We are not interested in information about a child once the parent leaves the company.
5.	Consider the Bank database given below. The primary keys are underlined. Assume relevant data types for attributes BRANCH (<u>branch_name</u>, branch_city, assets) CUSTOMER (<u>customer_name</u>, customer_street, customer_city) ACCOUNT (<u>account_number</u>, branch_name, amount) LOAN (<u>loan_number</u>, branch_name, amount) DEPOSITOR (<u>customer_name</u>, <u>account_number</u>) BORROWER (<u>customer_name</u>, <u>loan_number</u>) EMPLOYEE (<u>employee_name</u>, branch_name, salary) Creation of table with the appropriate constraints, CREATE, ALTER and DROP commands in SQL.

6.	Consider the Bank database given below. The primary keys are underlined. Assume relevant data types for attributes BRANCH (branch_name, branch_city, <u>assets</u>) CUSTOMER (customer_name, customer_street, customer_city) ACCOUNT (account_number, branch_name, amount) LOAN (loan_number, branch_name, amount) DEPOSITOR (customer_name, account_number) BORROWER (customer_name, loan_number) EMPLOYEE (employee_name, branch_name, salary) Insertion of tuple in each of the table (INSERT Command).
7.	Consider the Bank database given below. The primary keys are underlined. Assume relevant data types for attributes BRANCH (branch_name, branch_city, <u>assets</u>) CUSTOMER (customer_name, customer_street, customer_city) ACCOUNT (account_number, branch_name, amount) LOAN (loan_number, branch_name, amount) DEPOSITOR (customer_name, account_number) BORROWER (customer_name, loan_number) EMPLOYEE (employee_name, branch_name, salary) Insertion of tuple in each of the table (DELETE and UPDATE Command).
8.	Consider the Employee database given below. The primary keys are underlined. Assume relevant data types for attributes. EMPLOYEE (employee-name, <u>street</u>, <u>city</u>) WORKS (employee-name, company-name, salary) COMPANY (company-name, <u>city</u>) MANAGES (employee-name, manager-name) Create the above tables in SQL. Specify primary and foreign keys properly. Enter at least 5 tuples in each table with relevant data. Solve the following queries. - Find the names of all employees in the database who do not work for 'First Bank Corporation' - Find the names of all employees in the database who earn more than every employee of 'Small Bank Corporation'. Assume that all people work for at most one company. - Find the names, street address, and cities of residence for all employees who work for 'First Bank Corporation' and earn more than \$10,000 and less than \$20,000.

	iv. Find the names of all employees in the database who live in the same cities as the companies for which they work. v. Find the names of all employees in the database who earn more than every employee of 'Small Bank Corporation'. Assume that all people work for at most one company. vi. Find the names, street address, and cities of residence for all employees who work for 'First Bank Corporation' and earn more than \$10,000. vii. Find the names of all employees in the database who live in the same cities and on the same streets as do their managers.
9.	Consider the Order-Shipment database given below. The primary keys are underlined. Assume relevant data types for attributes. CUSTOMER (<u>cust #</u>, cname, city) ORDER (<u>order #</u>, odate, cust #, ord-Amt) ORDER – ITEM (<u>order #</u>, Item #, qty) ITEM (<u>item #</u>, unit price) SHIPMENT (<u>order #</u>, warehouse#, ship-date) WAREHOUSE (<u>warehouse #</u>, city) Create the above tables in SQL. Specify primary and foreign keys properly. Enter at least 5 tuples in each table with relevant data. Solve the following queries. i. List the no. of order placed by customer no. 5. ii. List customer details who have the largest order amount. iii. List the names of customers who have ordered at least 5 items. iv. List item numbers and their quantity for order number 5.
10.	Consider the Bank database given below. The primary keys are underlined. Assume relevant data types for attributes. BRANCH (<u>branch_name</u>, branch_city, assets) CUSTOMER (<u>customer_name</u>, customer_street, customer_city) ACCOUNT (<u>account_number</u>, branch_name, amount) LOAN (<u>loan_number</u>, branch_name, amount) DEPOSITOR (<u>customer_name</u>, account_number) BORROWER (<u>customer_name</u>, loan_number) EMPLOYEE (<u>employee_name</u>, branch_name, salary) Create the above tables in SQL. Specify primary and foreign keys properly. Enter at least 5 tuples in each table with relevant data. Solve the following queries. i. Find the names of all customers whose balance is less than 500. ii. Find all employees whose salary is greater than 1400 and working branch is not 'Downtown' iii. Calculate the average salary of all employees and display the

	average salary as “Avg_Salary” iv. Find the names of all customers whose city is not Brooklyn. v. Find the names of all customers who have taken loans. vi. Display all account numbers, branch name and corresponding branch city. vii. Display all loan numbers sorted by branch. viii. Display the names of Employees who earn the maximum salary.
11.	Consider the Order-Shipment database given below. The primary keys are underlined. Assume relevant data types for attributes. CUSTOMER (<u>cust #</u>, cname, city) ORDER (<u>order #</u>, odate, cust #, ord-Amt) ORDER – ITEM (<u>order #</u>, Item #, qty) ITEM (<u>item #</u>, unit price) SHIPMENT (<u>order #</u>, warehouse#, ship-date) WAREHOUSE (<u>warehouse #</u>, city) Create the above tables in SQL. Specify primary and foreign keys properly. Enter at least 5 tuples in each table with relevant data. Solve the following queries. i. Display the average order amount for day wise orders. ii. List the number of orders placed by each customer. iii. List the number of orders placed by each customer. iv. List the customer names who have not ordered for item no. 5. v. List the names of customers who have ordered at least 5 items.
12.	Demonstration of creating and deleting views for the database given BRANCH (<u>branch_name</u>, branch_city, assets) CUSTOMER (<u>customer_name</u>, customer_street, customer_city) ACCOUNT (<u>account_number</u>, branch_name, amount) LOAN (<u>loan_number</u>, branch_name, amount) DEPOSITOR (<u>customer_name</u>, account_number) BORROWER (<u>customer_name</u>, loan_number) EMPLOYEE (<u>employee_name</u>, branch_name, salary)

Suggested Learning Resources

Reference Books:

1. Ramez Elmasri and Shamkant B. Navathe: Fundamentals of Database Systems, 7th Edition, Pearson, 2016.
2. Silberschatz, Korth and Sudharshan: Data base System Concepts, 7th Edition, Mc-GrawHill, 2021.

Course Outcomes (COs):

At the end of the course student will be able to:

1. Design Entity relationship model for the real world scenario using modern IDE tools .(PO1, 2, 3,5 and PSO- 2).
2. Create an Enterprise Application using CRUD operation and DDL commands PO-1, 2,3, 4, 5 and PSO-2).
3. Design and Develop a database application and perform operations Using SQL (PO-1,2,3, 4, 5, and PSO-1,2)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment Tools	Marks	Course Outcomes (COs) addressed
Lab Test-I	10	CO1, CO2
Lab Test-II	10	CO3
Weekly Evaluation + Lab Record	30	CO1, CO2, CO3
The Final CIE out of 50 Marks = Marks of Lab Record + Marks scored in Lab Test-I + Marks scored in Lab Test-II		
Semester End Examination (SEE)	50	CO1, CO2, CO3

ALGORITHMS LABORATORY	
Course Code: CIL47	Credits: 0:0:1
Pre – requisites: Nil	Contact Hours: 15P
Course Coordinator: Dr. Sini Anna Alex	

Course Contents

Design and Develop algorithms to implement the following concepts using C/C++/Java:

Sl. No.	Lab Course Contents
1.	Asymptotic bounds and functions: Best Case, Average Case and Worst-Case Complexity
2.	Sorting and searching algorithms with Brute Force technique
3.	Stable matching algorithm using Iterative Improvement
4.	Graph Traversal: Breadth first search
5.	Graph Traversal: Depth first search
6.	Divide and Conquer Technique: Merge sort algorithm
7.	Divide and Conquer Technique: Quick sort algorithm
8.	Minimum Spanning Tree using Greedy Technique: Prim's and Kruskal algorithm
9.	Single Source Shortest Path using Greedy Technique: Dijkstra's algorithm
10.	Greedy approach: Job Sequencing with deadlines
11.	Dynamic Programming approach: Knapsack problem
12.	Branch and Bound Technique: Travelling Salesman Problem

Lab experiments will be scenario based implementation for the algorithm techniques specified in the course contents.

Suggested Learning Resources

Reference Books:

1. Algorithm Design - Jon Kleinberg and Eva Tardos, Tsinghua University Press (2005).
2. Anany Levitin: Introduction to the Design & Analysis of Algorithms, 2nd Edition, Pearson Education, 2007.

Course Outcomes (COs):

At the end of the course, the student will be able to:

1. Define the basic concepts and analyze worst-case running times of algorithms using asymptotic analysis. (PO-1,2,3,4, PSO-1,2)
2. Recognize the design techniques for graph traversal, divide and conquer, greedy and dynamic programming paradigm using representative algorithms. (PO-1,2,3, 4,5, 7, PSO-1,2,3)
3. Illustrate Branch and bound paradigm through NP complete problems. (PO-1, 3, 4, PSO-1,2)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment Tools	Marks	Course Outcomes (COs) addressed
Lab Test-I	10	CO1, CO2
Lab Test-II	10	CO3
Weekly Evaluation+Lab Record	30	CO1, CO2, CO3
The Final CIE out of 50 Marks = Marks of Lab Record + Marks scored in Lab Test-I + Marks scored in Lab Test-II		
Semester End Examination (SEE)	50	CO1, CO2, CO3

DATA VISUALIZATION LABORATORY

Course Code: CIL48

Credits: 0:0:1

Pre – requisites: Nil

Contact Hours: 15P

Course Coordinator: Dr. Siddesh G M/Mr. Subash N

Sl. No.	Lab Course Contents
	PART A (Data Visualization with Python)
1.	Write a program to create a line chart and a bar chart using Matplotlib.
2.	Write a program to create a scatter plot and a histogram using Seaborn.
3.	Write a program to create a boxplot and a violin plot using Seaborn.
4.	Write a program to generate a heatmap of a correlation matrix using Seaborn.
5.	Write a program to create a time series visualization using Pandas and Matplotlib.
6.	Write a program to create an interactive dashboard using Plotly.
	PART B: (Data Visualization with Power BI)
1.	Write a program to import a dataset and create a basic bar chart in Power BI.
2.	Write a program to create a pie chart and donut chart to represent categorical data distribution.
3.	Write a program to create line and area charts to visualize time-based trends.
4.	Write a program to use a tree map and funnel chart for hierarchical and stage-wise data visualization.
5.	Write a program to create a map visualization for geographical data using Power BI
6.	Write a program to design an interactive dashboard with multiple visuals and slicers.

Suggested Learning Resources

Reference Books:

1. Seema Acharya and Subhashini C. (2016). Data science and big data analytics: Discovering, analyzing, visualizing, and presenting data (1st ed.). Wiley.
2. Data Analysis with Microsoft Power Bi [ebook ed.] 1260458628, 9781260458626 <https://app.powerbi.com/>

Course Outcomes (COs):

At the end of the course, the student will be able to:

1. Data Visualization with Python Create static and interactive visualizations using Python (Matplotlib, Seaborn, Plotly). PO (1, 2, 3, 5) PSO (1, 2)
2. Data Visualization with Power BI Import datasets and create dynamic visualizations (charts, maps, dashboards) in Power BI. PO (1, 2, 3, 5) PSO (1, 2)
3. Interactive Dashboards and Advanced Visualizations Design interactive dashboards and advanced visualizations in Python and Power BI to communicate data insights. PO (1, 2, 3, 5) PSO (1, 2)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment Tools	Marks	Course Outcomes (COs) addressed
Lab Test-I	10	CO1, CO2
Lab Test-II	10	CO3
Weekly Evaluation+Lab Record	30	CO1, CO2, CO3
The Final CIE out of 50 Marks = Marks of Lab Record + Marks scored in Lab Test-I + Marks scored in Lab Test-II		
Semester End Examination (SEE)	50	CO1, CO2, CO3

NOSQL Databases	
Course Code: CIAEC49	Credits: 1:0:0:1
Pre – requisites: Nil	Contact Hours: 15L+15S
Course Coordinator: Mr. Subash N	

Course Contents

Unit I

Introduction: Why NoSQL? The Value of Relational Databases, RDBMS vs NoSQL, CAP Theorem, Impedance Mismatch, Application and Integration Databases, Attack of the Clusters, The Emergence of NoSQL.

Unit II

Introduction and CRUD Operations in MongoDB: MongoDB Datatypes, Where Clause, AND, OR operations in MongoDB, Insert, Query, Update, Delete, Remove and Projection, limit, skip and find method. Query selectors: comparison selectors, logical selectors, Geospatial selectors, Sort method, Aggregation functions

Unit III

Hbase: Introduction, Architecture, Hbase vs RDBMS, Shell Commands, Create Table, Insert & Retrieve Data, Performance Bottlenecks **Hive:** Introduction, Data types, Create & Drop Database, Create/Alter/Drop Table, Static Partitioning, Dynamic Partitioning, HiveQL: Operators, Functions, Group by and Having, Order By, Sort By, Joins

Unit IV

Cassandra: Architecture & Replication Factor Strategy, Features, Cassandra Key space: Create, Alter & Drop Key space in Cassandra. Cassandra Table — Create, Alter, Drop & Truncate. **Redis:** Introduction, Features, Datatypes, Keys, String.

Unit V

Graph Databases: Introduction, Features, Graph data in a relational database, querying graph data using MySQL, Neo4j in NoSQL space, Key-value stores, Column-family stores Data modeling in Neo4j, Neo4j CQL Read/Write Clauses: Set Clause, delete clause, Remove clause, Match clause, Where clause, Count Function, Order by clause, Limit, skip, String Functions, Aggregate Function.

Text Book:

1. Manu Sharma, “MongoDB Complete Guide”, 1e, bpb publication, 2023.
2. Edward Capriolo, Dean Wampler, Jason Rutherglen, "Programming Hive", O'Reilly Media, Inc., 2012.
3. Jeff Carpenter and Eben Hewitt, “Cassandra: The Definitive Guide”, 3rd Edition, O'Reilly Media, Inc., April 2020.
4. Eric Redmond and Jim R. Wilson, "Seven Databases in Seven Weeks: A Guide to Modern Databases and the Nosql Movement", O'Reilly; 1st edition, 2012.

References:

1. Kristina chodorow, “MongoDB: The Definitive Guide”, 2e, O'REILLY, 2013.
2. Kyle Banker, Peter Bakkum, Shaun Verch, Doug Garrett and Tim Hawkins, “MongoDB in Action” by 2e, Manning publication, 2016.
3. Edward Capriolo, Dean Wampler and Jason Rutherglen, Programming Hive, O'Reilly Media, Inc., 2012.
4. <https://neo4j.com/docs/cypher-manual>.

Course Outcomes (COs):

At the end of the course, students should be able to:

1. Understand the fundamentals of NoSQL databases and their comparison with RDBMS. **PO** (1, 2, 5), **PSO** (1, 2)
2. Gain proficiency in performing CRUD operations and queries in MongoDB. **PO** (1, 2, 3), **PSO** (1, 2)
3. Understand and apply the principles of HBase and Hive for large-scale data storage and retrieval. **PO** (1, 2,3, 5), **PSO** (1, 2)
4. Gain proficiency in working with Cassandra, focusing on architecture, key space, and table operations. **PO** (1, 2, 3), **PSO** (1, 2)
5. Learn and implement Graph Databases using Neo4j and understand the key-value and column-family store concepts. **PO** (1, 2, 3), **PSO** (1, 2)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment Tools	Marks	Course Outcomes (COs) addressed
Internal Test-I (CIE-I)	30	CO1, CO2
Internal Test-II (CIE-II)	30	CO3, CO4, CO5
Average of the two CIE shall be taken for 30 marks		
Other Components		
Case study	10	CO1, CO2, CO3
Assignment	10	CO4, CO5
The Final CIE out of 50 Marks = Average of two CIE tests for 30 Marks+ Marks scored in Case Study +Marks scored in Assignment		
Semester End Examination (SEE)	50	CO1, CO2, CO3 CO4, CO5